

Milo Walder

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EDUCATION

University of Maryland

B.S., Mechanical Engineering (GPA: 4.00)

College Park, MD

Expected May 2029

TECHNICAL EXPERIENCE

Terps Racing Baja

Powertrain Subteam Member

College Park, MD

Sept. 2025 – Present

- Redesigned Continuously Variable Transmission cover addressing security of attachment, vibrational damping, and interfacing with cooling components through iterative prototyping and decision matrices
- Manufactured two-part 3D-printed mold and hand-laid a carbon fiber-Kevlar composite cover, ultimately surviving a 4-hour endurance race and reducing weight by 25%
- Revamped front shaft bearing supports using SolidWorks topology optimization, correcting a bearing-load miscalculation and a critical frame-interface error before fabrication
- Eliminated a welded steel subframe by integrating supports into a CNC-machined design, reducing assembly weight 55%

Engineers Without Borders (UMD Chapter)

Nakifuma Subteam Member

College Park, MD

Sept. 2025 – Present

- Research anti-erosion pathway solutions for a school in rural Uganda, evaluating alternatives including stone pavers, geocell, and compacted aggregate based on cost and local material availability
- Synthesize input from community members, in-country experts, and a licensed PE mentor to inform pathway specifications for an upcoming 275 m² construction project

Risk and Decision Analysis Laboratory

Research Assistant

College Park, MD

Jan. 2026 – May 2026

- Supported graduate research applying dynamic probabilistic risk assessment (DPRA) and machine learning surrogate models to nuclear power plant risk assessment code
- Acquired working knowledge of industry grade nuclear modeling simulation code (ASYST/RELAP)

UMD Over Terrain Vehicle Project

Electronics and Coding Lead

College Park, MD

Sept. 2025 – Dec. 2025

- Led team of 8 students to design, build, and test an autonomous robot for obstacle navigation
- Designed and assembled electromechanical systems, integrating sensors, motors, and Arduino microcontrollers
- Programmed navigation algorithms in C++ for autonomous sensing and control

SKILLS

CAD: SolidWorks, Autodesk Inventor, Onshape, FEA Simulation

Programming: Java, MATLAB, Python, Arduino, EES

Engineering: Lathe, Mill, Composite Manufacturing, Laser Cutting, Rapid Prototyping, Breadboarding

Languages: Spanish (Conversational), English (Native)

AWARDS

- Dean's List, All Semesters
 - Korsnick Nuclear Engineering Innovation Award in Mechanical Engineering
 - University of Maryland President's Scholarship
- Awarded Aug. 2026
Awarded May 2025